COMPUTER PROJECT 1

Problem 1 : Newton basin in one dimension

Newton's method solves a nonlinear equation starting from an initial approximation. The method converges to a solution that is (surprisingly) dependent of that initial approximation. Two different starting points might produce sequences of approximations that converge to two different roots of the equation at hand. The set of points that will make the method converge to a given root is called the *basin of attraction* of that root. For this problem, you will classify a group of possible starting points for the algorithm by color, according to the basin they belong to.

Use Newton's method to solve the equation

$$x^3 - 6x^2 + 11x - 6 = 0$$

with tolerance 10^{-12}, 10 maximum number of iterations and starting points from 0 to 4 with step size 0.1, i.e., $0, 0.1, 0.2, ..., 3.9, 4$. The roots of the given equation are three:

$$x_1 = 1,\ x_2 = 2,\ x_3 = 3$$

For each starting point `x0`, plot a dot at the point $(x0, 0)$ in red, green or blue color, if the method converged to x_1, x_2 or x_3 respectively. If your method fails or does not converge, plot a black diamond. Plot the function on the same axis in a solid black line. The resulting figure will look similar to Figure 1.

Note: the example in the figure is for a different equation. The one you will generate will look similar.

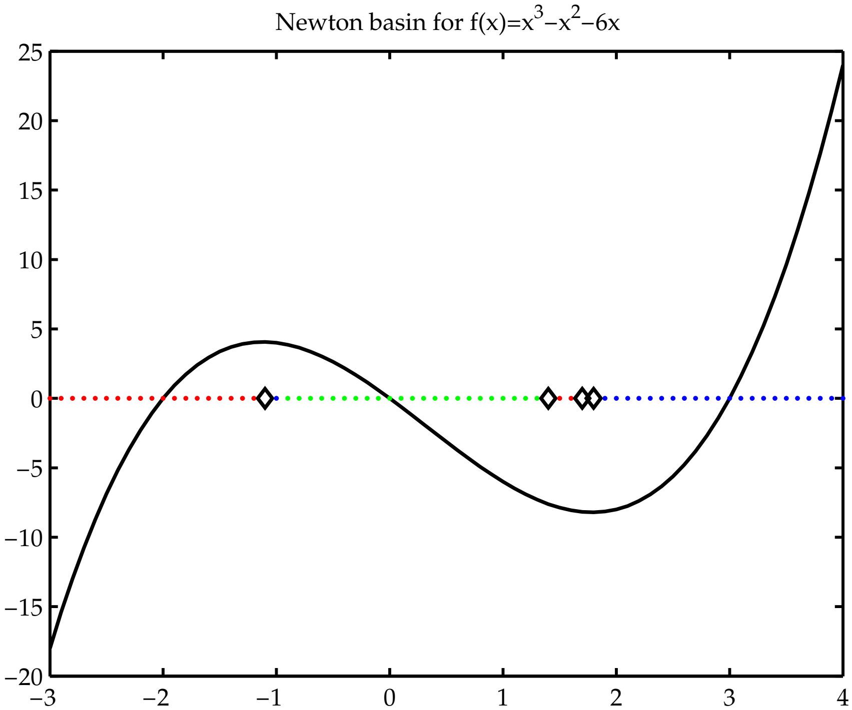

Figure 1: Example of a Newton basin for the equation $x^3 - x^2 - 6x = 0$ for initial approximations in $[-3, 4]$ with step 0.1, tolerance 10^{-12} and 10 maximum number of iterations. The roots of this equation are $-2, 0$ and 3. If the initial approximation of Newton's method is one of the red dots, the method will converge to -2; one of the green dots, will converge to 0; and one of the blue dots, will converge to 3. On the black diamonds, the method either did not converge in the maximum iterations given or it failed.